

Functional Annotation of *uppP* (Undecaprenyl-diphosphatase, UniProt Q88IY7) in *Pseudomonas putida* KT2440

1. Summary / Answer to the Research Question

The *uppP* gene of *Pseudomonas putida* KT2440 (locus **PP_2862**; synonyms *bacA*, *upk*; UniProt Q88IY7) encodes **undecaprenyl pyrophosphate phosphatase (UppP; EC 3.6.1.27)**, an integral **inner (cytoplasmic) membrane** enzyme of the **UppP/BacA family** (InterPro IPR003824; Pfam PF02673). Its primary, specific function is to **dephosphorylate undecaprenyl pyrophosphate (C55-PP) to undecaprenyl phosphate (C55-P)**, thereby regenerating the active mono-phosphorylated form of the **universal C55 isoprenoid lipid carrier ("bactoprenol")** that is essential for the trans-membrane assembly of peptidoglycan and other cell-envelope glycopolymers. It carries out this reaction **at the membrane-periplasm interface**, with its catalytic site facing the periplasm, and is proposed to help relocate the C55-P product back to the cytoplasmic face. Loss of this activity depletes the finite C55-P pool and arrests cell-wall synthesis; the enzyme's name *bacA* derives from the fact that maintaining this pool confers resistance to the antibiotic bacitracin.

Note on organism-specific evidence. The gene identity is unambiguous — the symbol *uppP*, the UppP/BacA domains, and the "undecaprenyl-diphosphatase / bacitracin resistance protein" description all match a single, well-defined bacterial enzyme family. However, **no primary experimental study of the *P. putida* KT2440 protein itself was found**; its annotation rests on the HAMAP rule (MF_01006) and very strong homology to biochemically characterized orthologs from *E. coli*, *Vibrio vulnificus*, and *Enterococcus faecalis*. The functional description below is therefore transferred from these orthologs by evolutionary/bioinformatic inference, which is highly reliable for this deeply conserved, single-function enzyme family.

2. Gene / Protein Identity Verification

Attribute	Value
UniProt accession	Q88IY7
Gene name / synonyms	<i>uppP</i> (<i>bacA</i> , <i>upk</i>)
Ordered locus	PP_2862
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125)
Enzyme	Undecaprenyl pyrophosphate phosphatase; EC 3.6.1.27
Family / domains	UppP family; UppP (IPR003824); BacA (PF02673)
Annotation basis	HAMAP-Rule MF_01006 (rule-based, homology inference)

The gene symbol, family, domains, and enzyme description are internally consistent and correspond to a single well-characterized enzyme. **The identification is confirmed.**

3. Primary Molecular Function — the catalyzed reaction and substrate specificity

UppP catalyzes:

undecaprenyl diphosphate (C55-PP) + H₂O → undecaprenyl phosphate (C55-P) + P_i (EC 3.6.1.27)

This activity was first established biochemically for the *E. coli* BacA/UppP protein and confirmed for multiple orthologs. Chang et al. describe the enzyme directly: "*Undecaprenyl pyrophosphate phosphatase (UppP), a cell membrane integral enzyme, catalyzes the dephosphorylation of undecaprenyl pyrophosphate to undecaprenyl phosphate, which is an essential carrier lipid in bacterial cell wall synthesis*" (

P 28302513

El Ghachi et al. state that *"the enzyme function is to synthesize the carrier lipid undecaprenyl phosphate that is essential for the biosynthesis of peptidoglycan and other cell wall components"* (P 15778224).

Substrate specificity. The physiological substrate is the fully-unsaturated C55 polyprenyl pyrophosphate. The curated UniProt/Rhea reaction for Q88IY7 specifies the exact stereochemistry — **di-trans,octa-cis-undecaprenyl diphosphate + H₂O → di-trans,octa-cis-undecaprenyl phosphate + phosphate + H⁺** (EC 3.6.1.27) — i.e. the bacterial C55 bactoprenol, not the α -saturated dolichol used by eukaryotes. This isoprenoid selectivity is consistent with the bacterial lipid-carrier machinery's strict preference for the fully unsaturated C55 species (context from

(P 9134438). The enzyme requires a **divalent metal cofactor (Mg²⁺ or Ca²⁺)** for activity (P 24855653).

4. Catalytic Mechanism and Active Site

UppP is a **metal-dependent, membrane-integral pyrophosphatase**. Two conserved sequence regions constitute the catalytic center: a glutamate-rich **(E/Q)XXXE** motif together with a **PGXSRXXT** motif and a conserved **His** residue. Mutagenesis in *E. coli* showed that E17A/E21A, H30A, S173A, R174A and T178A abolish activity, and that catalysis has *"an absolute requirement of magnesium or calcium ions"* (P 24855653).

Independent work on *V. vulnificus* UppP confirmed that *"the proposed catalytic residues Gln-13, Glu-17, His-26 and Arg-166 are directly involved in enzyme catalysis"*, with additional conserved Ser/Gly residues essential for activity (P 28302513).

The 2.0 Å *E. coli* crystal structure revealed *"the mechanistic basis for intramembranal phosphatase action and substrate specificity using an inverted topology repeat"* (P 29559664).

These catalytic residues are conserved in the *P. putida* ortholog under the HAMAP rule, supporting an identical mechanism.

5. Subcellular Localization and Membrane Topology

UppP is a **polytopic integral protein of the cytoplasmic (inner) membrane**, containing **6–8 transmembrane helices** depending on the prediction/structure method (UniProt annotates 6 for Q88IY7; the *E. coli* crystal structure resolves 8; *E. faecalis* UppP was "*predicted to be a hydrophobic protein with eight transmembrane helices*",

(P 23460607)

The **catalytic site faces the periplasm**: topological modeling places the two conserved catalytic motifs "*near the aqueous interface of UppP ... face the periplasm, implicating that its enzymatic function is on the outer side of the plasma membrane*"

(P 24855653)

This orientation is logical because the C55-PP substrate is generated on the periplasmic face after glycan polymers are transferred to their acceptors. The crystal structure further notes "*key structural motifs common to a variety of cross-membrane transporters,*" hinting at "*a potential flippase function in the specific relocation of the C55-P product back to the cytosolic space*"

(P 29559664)

— i.e., the enzyme may couple dephosphorylation to returning the carrier lipid to the cytoplasmic leaflet.

5b. Organism-Specific Sequence Verification (this study)

To confirm that the functional transfer applies to the *P. putida* protein itself and not merely to the family, the Q88IY7 sequence (277 aa; UniProt entry UPPP_PSEPK, PE=3) was retrieved and scanned for the UppP catalytic signatures defined by ortholog mutagenesis. **All experimentally validated catalytic residues are conserved at positions essentially identical to *E. coli*:**

Catalytic element	<i>E. coli</i> (validated)	<i>P. putida</i> Q88IY7	Status
N-terminal acidic motif (EGLTEFLP), catalytic Glu pair	E17, E21	E17, E21	conserved (motif at 17–24)
Conserved catalytic His	H30	H30 (G-H-Q)	conserved
PGXSRS signature, catalytic Ser/ Arg cluster	S173, R174, T178	P163-G-T-S166-R167-S168	conserved

Every residue shown by mutagenesis (E17A/E21A, H30A, S173A, R174A; *V. vulnificus* Gln13/Glu17/His26/Arg166) to be essential in orthologs

([P 24855653](#));

([P 28302513](#))

has an exact counterpart in Q88IY7. The protein is 56% hydrophobic (A/I/L/M/F/V/W), consistent with the polytopic ~8-TM inner-membrane topology of the BacA/UppP family. This provides **direct, protein-specific bioinformatic evidence** that Q88IY7 is a catalytically competent undecaprenyl pyrophosphate phosphatase, strengthening the HAMAP-rule annotation beyond mere family membership.

5c. Curated Database Annotation (UniProt Q88IY7)

The curated UniProt entry (UPPP_PSEPK) independently corroborates the literature-based conclusions specifically for this protein:

- **Function:** "*Catalyzes the dephosphorylation of undecaprenyl diphosphate (UPP). Confers resistance to bacitracin.*"
- **Subcellular location:** Cell inner membrane, with 6 annotated helical transmembrane segments (aa 44–64, 86–106, 110–130, 184–204, 215–235, 250–270) — a multi-pass polytopic protein (the *E. coli* crystal structure resolves 8 TM helices; both agree the enzyme is polytopic).
- **GO annotations:** molecular function *undecaprenyl-diphosphatase activity* (GO:0050380); cellular component *plasma membrane* (GO:0005886); biological process *peptidoglycan biosynthetic process* (GO:0009252), *cell wall organization* (GO:0071555), *regulation of cell shape* (GO:0008360), and *response to antibiotic* (GO:0046677).

These curated assignments match the experimental and structural evidence from orthologs, reinforcing confidence in the functional transfer.

6. Pathway Context — the Bactoprenol (C55-P) Recycling Cycle

UppP closes the **lipid-carrier recycling loop** central to cell-envelope biogenesis (

P 29559664

1. In the cytoplasm, peptidoglycan (and other glycan) subunits are assembled on **C55-P** (forming lipid I / lipid II, etc.).
2. The lipid-linked intermediate is **flipped to the periplasmic face**.
3. The glycan is **polymerized/transferred** onto the growing sacculus (or O-antigen, capsule, etc.), **releasing C55-PP** as a by-product.
4. **UppP dephosphorylates C55-PP → C55-P**, regenerating the active carrier, which returns to the cytoplasmic face to begin another round.

Because the C55-P pool is small and shared, this recycling step supplies the carrier for **multiple competing envelope pathways**: *"UndP ... ferries most glycans and glycopolymers across the cytoplasmic membrane ... In addition to peptidoglycan precursors, UndP transports O-antigen, capsule, wall teichoic acids, and sugar modifications"* (

P 39878533

In Gram-negative *P. putida*, the principal downstream demands are peptidoglycan synthesis and LPS **O-antigen** synthesis.

Two sources of substrate; one committed activation step. The authoritative review by Manat et al. (2014) frames UppP's position precisely: *"Undecaprenyl-phosphate originates from the dephosphorylation of its precursor undecaprenyl-diphosphate, with itself generated by de novo synthesis or by recycling after the final glycan transfer"* (

P 24799078

The Und-PP substrate therefore arrives from **two sources** — (i) *de novo* synthesis by the cytosolic *cis*-prenyltransferase undecaprenyl-diphosphate synthase (UppS/IspU), and (ii) **recycling** of C55-PP released after glycan transfer — and both converge on the UppP-catalyzed dephosphorylation that yields the usable Und-P carrier. This makes UppP the **committed activation step** of the bactoprenol cycle.

A second, unrelated phosphatase family explains redundancy. The same review notes that "two unrelated families of integral membrane proteins exhibit undecaprenyl-diphosphate phosphatase activity: *BacA* and members of the phosphatidic acid phosphatase type 2 super-family"

(
P 24799078
).

P. putida UppP (Q88IY7) belongs to the **BacA** family; PAP2-superfamily paralogs (homologs of *E. coli* PgpB/YbjG/LpxT) are expected to provide overlapping activity, which is why single *bacA*/*uppP* loss is typically sub-lethal even though the combined activity is essential.

7. Physiological Importance and Redundancy

Blocking C55-P regeneration is lethal: interruption "leads to the buildup of cell wall intermediates and cell lysis"

(
P 29559664
).

In *E. coli*, *bacA* alone is **not essential** because three additional membrane phosphatases (PAP2-family YbjG, PgpB, and YeiU/LpxT) provide overlapping Und-PP-phosphatase activity; only a conditional *bacA ybjG pgpB* triple mutant "accumulate[s] soluble peptidoglycan nucleotide precursors and ... lyse[s] ... due to the depletion of the pool of undecaprenyl phosphate and consequent arrest of cell wall synthesis"

(
P 15778224
).

BacA/UppP contributes the largest single fraction of this activity. The degree of redundancy in *P. putida* has not been experimentally mapped, but analogous PAP2-type phosphatases are encoded in its genome, so *uppP* is expected to be the major but not sole contributor.

8. Secondary Phenotypes that Illuminate the Precise Role

- **Bacitracin resistance (the *bacA* name).** Bacitracin acts by sequestering C55-PP, starving the cell of recyclable carrier. Elevated UppP activity replenishes C55-P and confers resistance: overexpression "result[s] in bacitracin resistance and increased levels of undecaprenyl pyrophosphate phosphatase activity"

(
P 15778224
),

and in *E. faecalis* deletion lowered the bacitracin MIC from 32–48 to 3–6 mg/L while

overexpression raised it to ≥ 256 mg/L, with MICs of unrelated antibiotics unchanged

([P 23460607](#)).

This specificity confirms that the phenotype flows directly from C55-PP $\rightarrow$ C55-P turnover, not a general stress effect.

- **Bacteriocin receptor.** UppP serves as the membrane receptor for two-peptide (class IIb) bacteriocins: resistance mutations to lactococcin G all mapped to "*the gene uppP (bacA), encoding an undecaprenyl pyrophosphate phosphatase; a membrane protein involved in peptidoglycan synthesis*"

([P 24779486](#)).

This reflects the enzyme's exposed, essential position in the membrane rather than a distinct biochemical activity.

These are **downstream consequences** of the single core function and should not be mistaken for the primary role.

9. Supported and Refuted Hypotheses

Supported - H1: Q88IY7 is an undecaprenyl pyrophosphate phosphatase (EC 3.6.1.27) — supported by family/HAMAP assignment and biochemistry of orthologs. - H2: The enzyme functions in the inner membrane with a periplasm-facing active site — supported by topology, mutagenesis, and crystal structure of orthologs. - H3: Its biological role is bactoprenol (C55-P) recycling for peptidoglycan/envelope glycopolymer synthesis — supported by genetics and pathway data. - H4: Catalysis is Mg^{2+}/Ca^{2+} -dependent with conserved Glu/His/Arg active-site residues — supported by mutagenesis.

Refuted / not supported - The enzyme is *not* a soluble cytoplasmic phosphatase, and its function is *not* primarily a stress/resistance determinant; bacitracin resistance and bacteriocin sensitivity are secondary to lipid-carrier recycling.

10. Limitations and Future Directions

- **No direct experimental data on the *P. putida* KT2440 protein;** all mechanistic detail is transferred from orthologs. Direct enzymatic assay of purified PP_2862, and a $\Delta uppP$ phenotype in *P. putida*, would confirm the transfer and quantify redundancy with PAP2-type phosphatases.

- The proposed **flippase/product-relocation activity** remains a structural inference and is unproven for any ortholog.
- The relative contribution of *uppP* vs. PAP2-family paralogs to total Und-PP-phosphatase activity in *Pseudomonas* is unknown.

Key References

- Workman, Worrall & Strynadka (2018) *PNAS/JBC* — crystal structure of *E. coli* UppP.

P 29559664

- Chang et al. (2017) — *V. vulnificus* UppP characterization/mutagenesis.

P 28302513

- Chang et al. (2014) — proposed carrier-lipid binding site; topology; metal dependence.

P 24855653

- El Ghachi, Derbise, Bouhss & Mengin-Lecreulx (2005) *J. Biol. Chem.* — multiple Und-PP phosphatases; essentiality of the pathway.

P 15778224

- Shaaly, Kalamorz, Gebhard & Cook (2013) — UppP and bacitracin resistance in *E. faecalis*.

P 23460607

- Kjos et al. (2014) — UppP as bacteriocin receptor.

P 24779486

- Manat et al. (2014) *Microb. Drug Resist.* — authoritative review on Und-P metabolism; BacA vs PAP2 phosphatase families; de novo vs recycling.

P 24799078

- Roney & Rudner (2025) — UndP as universal carrier for multiple envelope pathways.

P 39878533

- Jorgenson (2026) — review: Und-P sequestration and the bacterial cell envelope.

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